

SOCIOLOGY 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: Who Gets Care?

Grading guide. Total 22 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Counts into rates

1. Three made-up countries. Work out each one's preterm RATE as a number out of 100 (a percent). Preterm $\div$ births $\times$ 100. (2 points per row: 1 for the division set up, 1 for the answer.) (6 pts)

Answer: A: $160 \div 1,000 = 0.16 \rightarrow 16$ per 100 (16%). B: $300 \div 3,000 = 0.10 \rightarrow 10$ per 100. C: $30 \div 500 = 0.06 \rightarrow 6$ per 100.

2. Which country has the MOST preterm babies, and which has the HIGHEST preterm rate? (1 pt)

Answer: A. B has the most babies (300); A has the highest rate (16 per 100). - Counts and rates answer different questions. B has more preterm babies only because it has more babies; A's babies are far more likely to be born early.

Part B — Per thousand

3. In one region, 5,000 babies were born last year and 28 died before their first birthday. What is the infant death rate per 1,000? (Deaths $\div$ births $\times$ 1,000.) Compare it with about 4.5 for white babies and about 10.9 for Black babies in the US: which is it closer to? (3 pts)

Answer: $28 \div 5,000 = 0.0056 \rightarrow 5.6$ per 1,000 (2 pts; 1 pt for the setup if the arithmetic slips). Closer to 4.5 — about the US overall rate (1 pt).

Part C — The map

Three towns. Riverton has a Level I nursery (healthy newborns only), 20 minutes from the Okafor family. Hillsdale has a Level II unit (a little early, a little sick), 1 hour away. The city has a Level IV NICU (everything, including surgery), 3 hours away. Mrs. Okafor is 27 weeks pregnant and her doctor thinks the baby may come in the next few days.

4. Using regionalization from last week, where should Mrs. Okafor be – and when should she go? What is the danger of waiting until the baby is born? What could the region do so families like hers don't face a 3-hour drive? (3 pts)

Answer: To the city's Level IV NICU, NOW, before the birth – move the mother, not the baby (1 pt). Waiting means a 27-week baby born at a Level I nursery with no ventilator or NICU team, then a 3-hour transport in a traveling isolette – the most dangerous hours of its life (1 pt). Fixes: a Level III unit in Hillsdale, a transport team and helicopter, a nurse visiting high-risk mothers at home, clinics that catch early labor sooner (1 pt for any sensible idea).

Part D — Social determinants

5. Draw a line from each family's situation to the social determinant it is. (4 pts)

Answer: No hospital within two hours -> Distance to care | Skipping meals to pay rent -> Money and food | Feeling ignored at every clinic visit -> Being listened to (respect) | Working nights and days with no rest -> Stress (1 pt each)

Part E — Flashbacks

6. From NICU Tutor: where incubators are scarce, what do doctors teach instead, and why does it work? (1 pt)

Answer: Kangaroo care – skin-to-skin on a parent's chest. A parent's body is a warming machine that's always available, and it steadies heart rate and breathing.

7. From last week's Sociology: which level of newborn unit does complex surgery? (1 pt)

Answer: Level IV – the regional center.

Part F — Think like a doctor

8. You run a small hospital in a poor country. Tonight you have two working incubators and ten premature babies. Using Psychology (last week and today's flashback), Physics (bubble CPAP from Week 1), and today's Sociology, what would you do – tonight, and next year? (3 pts)

Answer: Tonight: incubators for the two sickest or smallest; kangaroo care for the others (the Bogotá idea – warmth, steady breathing, fewer deaths, no machine needed); plastic wraps and hats for warmth; a bubble CPAP made from tubing and a bottle of water for a baby working to breathe (1 pt for using KMC; 1 pt for any other basic-care idea). Next year: train every nurse and mother in kangaroo care, count your babies and your deaths (rates!) so you can show what's working, ask for what the numbers say you need – and put a clinic where mothers can reach it before birth (1 pt).

HOME LAB (optional, not graded): Fieldwork: how far is your NICU?

You need: a grown-up, a map app or website, and about ten minutes

1. With a grown-up, find the nearest hospital to your home that has a NICU (try searching 'Level III NICU near me'; hospital websites usually say their level).
2. Write its name, the distance, and the drive time.
3. Now find the nearest hospital that delivers babies at all. Distance and time?
4. Imagine your family lived 2 hours from both. Write one thing that would change about a preterm birth.
5. Bonus: ask a friend or cousin in another town the same question and compare.

What you should see: You'll have two distances and two drive times — most city families are under 30 minutes from a Level III unit; many rural families are over an hour from any birth care at all. The point is the comparison: the same baby, different distances, different odds. That's a map of who gets care.